

KESWICK, United Kingdom

WMO: 032120

Lat: 54.6139N

Lon: 3.1567W

Elev: 81

StdP: 100.36

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.5	-2.9	-6.1	2.3	-0.8	-4.8	2.5	-0.2	11.5	7.5	10.6	7.8	0.5	120	0.543

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	24.3	18.0	22.2	16.8	20.6	16.0	18.7	22.9	17.6	21.0	16.6	19.6	2.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.0	12.3	20.6	16.1	11.5	19.2	15.2	10.9	18.2	53.1	23.0	49.7	21.1	46.9	19.6	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.0	6.9	-7.0	27.1	2.2	2.0	-8.5	28.5	-9.8	29.7	-11.0	30.9	-12.6	32.3		
			-7.0	20.9	2.2	1.9	-8.6	22.2	-9.8	23.3	-11.1	24.3	-12.7	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	4.8	4.8	5.9	8.2	11.0	13.7	15.4	15.0	13.0	10.2	7.1	4.9
	DBStd	4.85	3.15	3.02	2.88	2.75	2.89	2.56	2.30	2.12	2.36	2.97	3.27	3.79
	HDD10.0	815	163	146	128	67	21	2	0	0	3	33	94	159
	HDD18.3	3218	421	378	384	304	227	142	96	106	159	251	336	416
	CDD10.0	650	1	1	3	13	53	111	168	156	94	41	8	2
	CDD18.3	12	0	0	0	0	1	2	6	3	1	0	0	0
	CDH23.3	102	0	0	0	0	4	22	52	23	1	0	0	0
	CDH26.7	11	0	0	0	0	0	3	5	3	0	0	0	0
Wind	WSAvg	2.5	3.5	3.4	3.3	2.8	2.4	1.9	1.6	1.6	1.8	2.3	2.8	3.2
Precipitation	PrecAvg	1415	150	113	98	86	85	84	97	110	117	156	156	166
	PrecMax	1790	237	275	173	140	162	226	198	224	176	277	399	468
	PrecMin	1011	11	28	31	35	21	30	52	18	23	43	81	28
	PrecStd	211	58	67	36	29	39	44	34	48	42	72	70	94
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	13.2	15.5	20.6	23.7	26.0	27.2	26.5	22.3	18.3	15.0	13.5
		MCWB	10.5	10.2	10.4	13.3	16.5	19.3	19.7	18.8	17.3	14.4	13.0	11.6
	2%	DB	11.2	11.4	12.7	16.7	21.1	22.8	24.7	22.7	20.0	16.5	13.8	12.2
		MCWB	9.5	9.3	9.7	11.5	15.2	17.2	18.2	17.5	15.8	14.0	11.9	10.5
	5%	DB	10.2	10.4	11.4	14.5	18.7	20.7	22.4	20.6	18.5	15.6	12.7	11.3
		MCWB	8.5	8.5	9.0	10.3	13.7	15.9	17.1	16.4	14.9	13.4	11.0	9.6
	10%	DB	9.4	9.4	10.4	12.8	16.4	18.8	20.3	19.0	17.2	14.7	11.7	10.3
		MCWB	7.8	7.6	8.2	9.5	12.2	14.6	16.2	15.5	14.2	12.5	10.2	8.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	10.8	11.6	13.8	17.5	19.8	20.8	19.5	18.1	15.6	13.5	12.0
		MCDB	12.1	12.3	14.1	18.4	22.5	24.3	26.1	24.3	21.1	17.0	14.7	13.2
	2%	WB	9.8	9.7	10.2	12.1	15.6	17.8	18.9	18.1	16.7	14.6	12.2	10.9
		MCDB	11.0	11.1	12.1	15.7	20.0	21.8	23.3	21.6	19.1	16.1	13.4	12.1
	5%	WB	8.8	8.7	9.3	11.0	14.1	16.3	17.6	17.0	15.5	13.7	11.3	9.9
		MCDB	10.0	10.2	11.0	13.8	17.9	19.9	21.2	19.6	17.6	15.4	12.5	11.2
	10%	WB	7.9	7.8	8.4	10.0	12.8	15.0	16.6	16.2	14.6	12.7	10.3	8.8
		MCDB	9.2	9.2	10.2	12.5	15.9	18.0	19.7	18.4	16.7	14.4	11.6	10.2
Mean Daily Temperature Range	5% DB	MDBR	4.7	5.4	6.3	7.5	8.3	8.0	7.9	7.4	7.3	6.2	5.3	5.4
		MCDBR	5.0	5.7	7.5	10.6	12.2	11.7	12.2	10.5	9.4	6.5	5.1	5.5
		MCWBR	4.8	4.9	5.3	6.6	7.5	7.3	6.9	6.3	5.9	5.0	4.6	5.1
	5% WB	MCDBR	4.9	5.2	6.5	8.9	11.2	10.6	10.5	8.9	7.8	5.9	5.0	5.4
		MCWBR	4.9	4.8	5.2	6.1	7.2	7.0	6.5	5.8	5.5	5.0	4.8	5.2
Clear-Sky Solar Irradiance	taub	0.325	0.336	0.363	0.382	0.386	0.383	0.384	0.371	0.359	0.344	0.320	0.311	
	taud	2.270	2.354	2.308	2.284	2.322	2.356	2.357	2.405	2.444	2.433	2.347	2.249	
	Ebn at Noon	601	746	805	841	859	864	855	845	808	730	611	529	
	Edh at Noon	59	76	99	116	118	115	113	102	86	70	54	49	
All-Sky Solar Radiation	RadAvg	0.55	1.22	2.33	3.72	4.84	4.87	4.70	3.79	2.68	1.44	0.70	0.42	
	RadStd	0.07	0.14	0.22	0.37	0.37	0.48	0.49	0.28	0.23	0.17	0.07	0.04	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (63 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page